Qualifying Image-Grounded Tag Retrieval in Piping and Instrumentation Diagrams with Source-Isolated Counterfactual Evaluation

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Abstract

Engineering use of vision–language model outputs requires a qualification decision, not only an aggregate benchmark score. We present SABER-PID, a source-isolated counterfactual workflow for determining whether candidate-tag retrieval from piping and instrumentation diagrams (P&IDs) is supported for a stated model, visual-input budget, task, and drawing family. On 100 unseen PIDQA source drawings, frozen Qwen3-VL-8B at a 3072-side, 192-token operating point produced 192 true positives, 152 false positives, and 156 false negatives: pooled precision 0.5581, recall 0.5517, and F1 0.5549; source-macro F1 was 0.5810. Source-shuffled and no-image controls reduced pooled F1 to 0.0062 and 0.0000. At a common 512-token output cap, the 3072-minus-768 visual-input-budget effect was +0.5250 (95% source-bootstrap interval [0.4224, 0.6293]). A frozen 54-cell extension across Qwen3-VL-8B/32B, InternVL3.5-8B, three pairwise source-disjoint PIDQA subsets, and two pre-existing prompts gave positive lower bounds for all 36 correct-minus- control F1 intervals; median correct-image F1 was 0.623, 0.651, and 0.012, respectively. Aggregate strict accuracy did not identify this capability: a task-prior diagnostic (0.3375) and no-image condition (0.3025) exceeded the correct-image aggregate score (0.2875). A frozen full-image OCR comparator reached pooled F1 0.5933. Prediction-only operating modes exposed complementary errors: union increased recall to 0.7040 and pooled/source-macro F1 to 0.6339/0.6602, whereas intersection yielded precision 0.9907 with interval [0.9681, 1.0000]. Set-B-excluded checks retained positive pooled union-minus-Qwen effects on two 83-source subsets (+0.0366 and +0.0440); stricter mutually disjoint 65-source checks showed +0.0325 and +0.0509, with one interval crossing zero. Accordingly, the supported outcome is bounded candidate-tag retrieval within PIDQA, with selectable coverage and precision modes; general P&ID understanding, topology reasoning, cross-model invariance, and real-plant deployment are not qualified.

Keywords: engineering document intelligence; vision–language models; optical character recognition; tag retrieval; source isolation; counterfactual evaluation

1 Introduction

Piping and instrumentation diagrams encode equipment, instruments, process lines, and control relationships in visually dense engineering documents. Their identifiers connect drawings to equipment lists, maintenance records, and engineering databases. Candidate-tag retrieval is therefore useful for document indexing, search, and migration even when full graph reconstruction is not yet available.

The central deployment question is not simply whether a multimodal model obtains a high score. A model can exploit answer priors, repeated drawing identity, or task regularities; small text may also disappear under an insufficient visual-input budget. In our benchmark, aggregate strict accuracy illustrates this problem directly: the task-prior diagnostic scores 0.3375, the no-image condition 0.3025, and the correct-image condition 0.2875. Aggregate accuracy therefore cannot by itself qualify image reading. A useful engineering evaluation must instead test whether a task-specific effect follows the requested drawing, survives a matched output budget, and remains within explicitly stated scope boundaries.

Prior P&ID work has emphasized symbol and text recognition, line extraction, graph conversion, and end-to-end digitization [1, 2, 3, 4]. PIDQA contributes 64,000 graph-derived questions from 500 synthetic source sheets [6]. This resource enables controlled multimodal evaluation but also creates dependence: many records share the same drawing. Question-random splitting can therefore expose source-specific information unavailable on an unseen drawing, a grouped-data failure addressed by multiple-source cross-validation and leakage-aware evaluation [7, 8].

Document VQA studies likewise distinguish answer similarity from document groundedness [9, 11]. Counterfactual VQA research uses altered visual or textual evidence to expose shortcut dependence [12]; large-VLM studies show that fluent predictions may remain unsupported by the visual input [13, 14]. These findings motivate an engineering qualification view: a capability is accepted only for the task, model, budget, and source family for which its evidence gates pass.

We call this workflow *SABER-PID*: source-isolated, answer-hidden, budget-recorded, evidence-counterfactual, and reported-by-task evaluation. It is an evaluation and decision workflow, not a new model architecture. The contributions are:

1. a source-level qualification contract that separates aggregate diagnostics from the candidate-tag retrieval claim;
2. paired correct-image, source-shuffled-image, no-image, and matched-budget contrasts on answer-hidden manifests;
3. pooled and source-macro estimators, source-cluster uncertainty, TP/FP/FN accounting, and per-drawing candidate workload;
4. a complete family of reference-free OCR–VLM operating modes plus overlap-audited frozen-rule checks that distinguish initial description from later stability evidence.

2 Related work and evaluation objective

2.1 P&ID recognition and engineering information extraction

Classical P&ID digitization pipelines decompose recognition into symbols, text, lines, associations, and graph construction [1, 2]. Kim et al. developed symbol and text recognition for high-density industrial diagrams [3]; Su et al. combined symbol, text, and pipeline recognition on high-resolution industrial drawings [4]. More recent work demonstrates why raw OCR is only an intermediate layer: engineering identifiers require format-aware validation and semantic interpretation [5]. Our target is narrower than end-to-end digitization. We evaluate retrieval of candidate value tags from a complete drawing and do not infer connectivity or topology.

2.2 Document VQA, grounding, and grouped evaluation

DocVQA and ChartQA established document-oriented question answering as a joint text, layout, and reasoning problem [9, 10]. Grounding-aware evaluation asks whether an answer can be attributed to the input document rather than only whether its surface form matches a reference [11]. Counterfactual samples and contrastive controls expose visual and linguistic shortcuts [12, 14]. Separately, multiple-source cross-validation requires related observations to remain in one partition [7]. SABER-PID combines these ideas as a measurement contract: drawings are the resampling and splitting unit, references are hidden from inference, and image identity is intervened on directly.

2.3 Scope of the present contribution

We do not train a detector, optimize a prompt family, or claim benchmark leadership. Frozen Qwen3-VL and InternVL checkpoints [16, 17], a frozen PaddleOCR pipeline [15], and deterministic set operations serve as instruments. The objective is to decide whether a practically useful candidate-tag capability is qualified under the frozen contract and to quantify the review burden of its operating modes.

3 Materials and methods

3.1 Dataset, source identity, and answer isolation

PIDQA contains 64,000 questions from 500 Dataset-PID sheets, with 16,000 questions in each of four task families: count, spatial count, connectivity, and value retrieval [6]. The source sheet is the clustering and splitting unit. Primary Set B contains one deterministically selected record per source and task for 100 source sheets (400 records); tag metrics use its 100 value records. Public identifiers are ranked by SHA-256. Answers, Cypher queries, and predictions never enter selection.

Every model-facing manifest contains only instance ID, source ID, task, question, image path, and public template fields. References are stored separately for scoring. An automated isolation audit covers seven public manifests and 13 raw output files from the primary controls and finds no `answer` or `cypher` field in model inputs. Immutable inputs and outputs are identified by SHA-256 in the release manifest.

3.2 Source-split diagnostic

For split seeds 3, 17, 29, 43, and 71, question-random and source-isolated partitions use 60% training, 20% calibration, and 20% test allocations. An input-retrieval-plus-prior diagnostic builds an unambiguous training cache keyed by exact source-image SHA-256 and a frozen semantic signature of public question fields. A hit returns the unique cached training answer; a miss uses the training-only task majority. The source identifier itself is not a key. Each split contains 38,400 training and 12,800 test records. This deterministic diagnostic quantifies an available same-drawing route; it is not presented as observed VLM behaviour.

3.3 Frozen inference conditions

The primary instrument is Qwen3-VL-8B-Instruct with the pre-specified P0 prompt and greedy decoding. It receives a maximum image side of 768 or 3072 and an output cap of 192 or 512 tokens, as labelled for each contrast. The matched-budget contrast fixes the output cap at 512. The source-shuffled condition applies a deterministic Sattolo derangement across the 100 drawings with no fixed point. The no-image condition retains model, prompt, question, decoder, and 192-token cap while removing image content. Two source partitions selected before their scores were read are retained as descriptive budget-sensitivity analyses.

The cross-model boundary check uses InternVL3.5-8B on the same 100 value records under correct-image, source-shuffled-image, and no-image conditions, seven realised 448-pixel tiles for image conditions, a 512-token cap, and the tokenizer's documented Mistral-regex correction. A frozen replication matrix then crosses Qwen3-VL-8B, Qwen3-VL-32B, and InternVL3.5-8B with Set B and two additional pairwise source-disjoint 65-drawing subsets, both pre-existing prompts, and correct, source-shuffled, and no-image conditions. Qwen uses a 3072-side/512-token configuration; InternVL uses at most 12 realised tiles and the same output cap. This yields 54 complete cells and 16,560 predictions. Both prompts are reported without best-prompt selection. The OCR comparator uses PaddleOCR 2.8.1 with PaddlePaddle 2.6.2, the English PP-OCRv3 detector/PP-OCRv4 recognizer, the complete image, no angle classifier, and detector maximum side 3072. A deterministic vertical prefix–suffix geometry join reconstructs split identifiers. The join uses coordinates and a legal tag grammar only; it does not consult any reference.

3.4 Reference-free OCR–VLM operating modes

Let Q_i and O_i denote Qwen and geometry-joined OCR candidate sets for source i. Four pre-declared, score-free operations are evaluated: $Q_i \cup O_i$, $Q_i \cap O_i$, OCR-if-nonempty with Qwen fallback, and Qwen-if-nonempty with OCR fallback. Union targets coverage, intersection targets a conservative shortlist, and fallback retains a single instrument's set when available. The complete rule family is always reported.

The evidence phases are separated in Table 1. Set B first described component complementarity and the rule family. The rules were then frozen before scoring seed-29 and seed-31 partitions. Because those partitions overlap Set B and each other, we additionally remove Set B sources and then construct mutually disjoint subsets. These checks measure within-family stability; they are not treated as independent external replication.

Table 1: Evidence phases, frozen decisions, and allowed interpretation.

Phase	Data role	Frozen before scoring	Allowed interpretation
Pilot/configuration	Earlier development records	Prompt, parser, legal tag grammar, visual-input conditions	Select a measurement configuration; no confirmatory claim
Primary qualification	Set B, 100 unseen sources	Source selection, Qwen condition, controls, estimators	Qualify or reject the bounded Qwen tag-retrieval claim
Cross-scale replication	Three pairwise source-disjoint subsets	Three models, two prompts, three evidence conditions, estimators	Test requested-drawing direction and magnitude stability within PIDQA
Descriptive fusion	Set B predictions	Component outputs; rule family defined after component inspection	Describe operating modes and complementarity
Frozen-rule transfer	Seed-29/31 predictions, with audited exclusions	All four set/fallback rules; no re-selection	Assess within-family direction and sensitivity, not external replication

3.5 Estimators, uncertainty, and workload

For source i, reference and prediction sets are T_i and P_i. Counts are $TP_i = |T_i \cap P_i|$, $FP_i = |P_i \setminus T_i|$, and $FN_i = |T_i \setminus P_i|$. Pooled (micro) estimators first sum counts across sources:

$$P_{\text{micro}} = \frac{\sum_i TP_i}{\sum_i (TP_i + FP_i)}, \quad R_{\text{micro}} = \frac{\sum_i TP_i}{\sum_i (TP_i + FN_i)}, \quad F1_{\text{micro}} = \frac{2\sum_i TP_i}{2\sum_i TP_i + \sum_i FP_i + \sum_i FN_i}.$$

Source-macro precision, recall, and F1 are the arithmetic means of the corresponding per-source values. If both reference and prediction are empty, per-source precision, recall, and F1 are 1; an empty prediction against a non-empty reference scores 0. Set B contains seven empty-reference sources, so we additionally report a non-empty-reference macro sensitivity. Exact-set accuracy is the fraction with normalized set equality.

All intervals are 95% percentile intervals from 10,000 source-cluster bootstrap replicates using fixed, comparison-specific PCG64 seeds. Paired method differences resample identical source indices. The correct-minus- source-shuffled contrast is the primary grounding contrast; correct-minus- no-image and matched-budget contrasts are secondary controls. Fusion comparisons and transfer checks are labelled descriptive. No equivalence margin is introduced after observing results.

Per-drawing workload is $|P_i|$, reported as median and interquartile range (IQR), together with false-candidate count $|P_i \setminus T_i|$. These values operationalize the number of candidates a deterministic downstream rule or engineering review stage would need to process.

4 Results

4.1 Aggregate performance is a boundary, not a qualification result

Figure 1 summarizes the decision. Aggregate strict accuracy is highest for the task-prior diagnostic (0.3375), followed by no image (0.3025) and the correctly paired image (0.2875). In contrast, the value-tag task passes the requested-drawing gate: pooled F1 is 0.5549 with the correct drawing, 0.0062 with a source-shuffled drawing, and 0.0000 without an image. It also passes the matched-budget gate: at a common 512-token cap, pooled F1 is 0.5253 at 3072-side and 0.0003 at 768-side. The qualified scope is therefore Qwen candidate-tag retrieval on PIDQA at the stated high-detail budget, not aggregate question answering or general P&ID understanding.

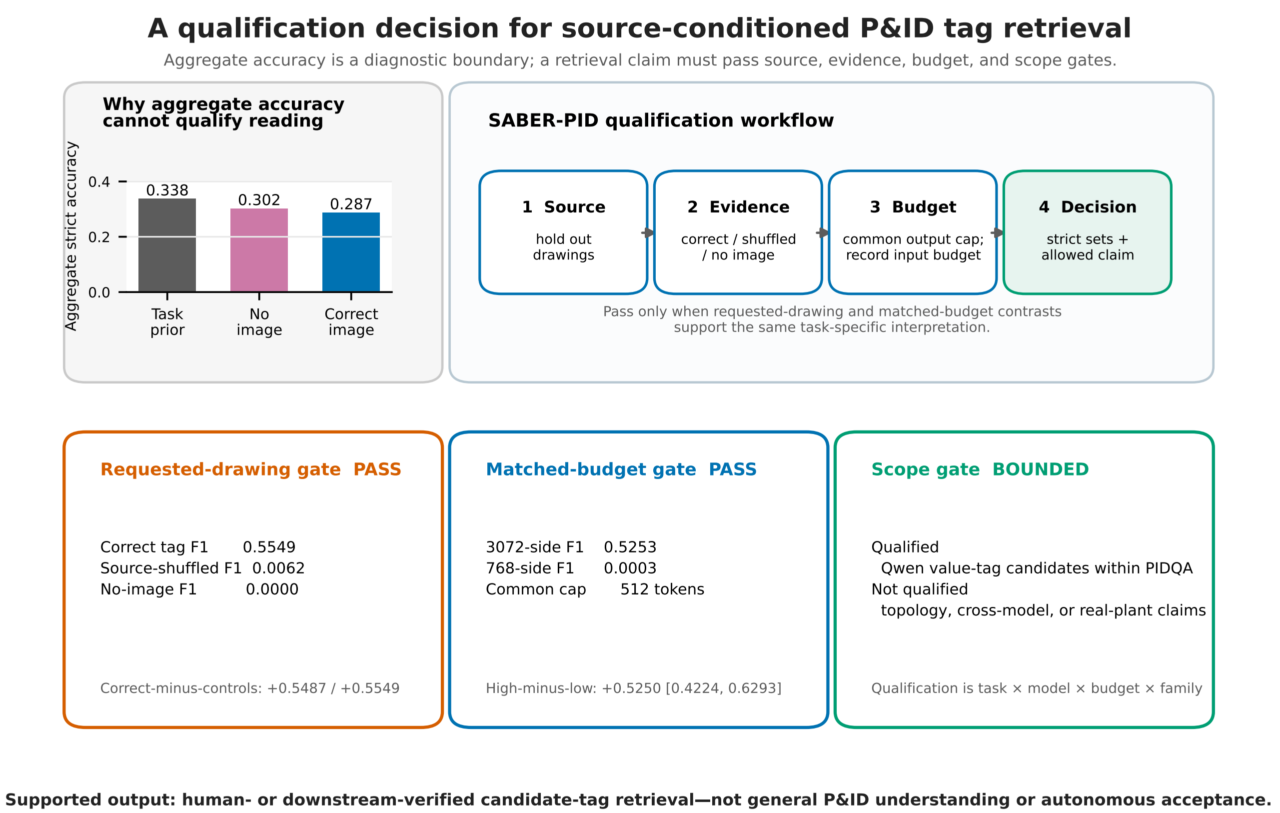

Figure 1: Qualification decision. Aggregate accuracy is retained as a diagnostic boundary. A task-specific claim proceeds through source, counterfactual-evidence, matched-budget, and scope gates. Values are generated from the frozen analysis artifacts.

Across the five split seeds, the input-retrieval-plus-prior diagnostic scores 50.9% under question-random splitting and 31.5% under source isolation, a mean difference of 19.5 percentage points (individual differences 13.6–21.3 points). Exact-image semantic-cache hits create the additional route only when the same drawing crosses the split. The result motivates drawing-level isolation without claiming that the evaluated VLM used that route.

4.2 The tag signal follows the requested drawing and visual-input budget

At 3072-side and 192 output tokens, Qwen produces 192 true positives, 152 false positives, and 156 false negatives. Pooled precision, recall, F1, and exact-set accuracy are 0.5581, 0.5517, 0.5549, and 0.1900. Source-macro F1 is 0.5810 ([0.5118, 0.6501]); excluding the seven empty-reference sources gives 0.6139. Correct minus source-shuffled pooled F1 is +0.5487 ([0.4729, 0.6239]), and correct minus no-image is +0.5549 ([0.4799, 0.6322]). Thus, the useful signal tracks the requested source drawing rather than the prompt or arbitrary P&ID pixels.

At a common 512-token cap, the high-minus-low visual-input-budget effect is +0.5250 ([0.4224, 0.6293]). At the original 192-token cap it is +0.5549 ([0.4774, 0.6314]). The seed-29 and seed-31 descriptive partitions yield +0.6783 ([0.6163, 0.7384]) and +0.6500 ([0.5684, 0.7295]). All four effects are positive, while only Set B carries the primary qualification role.

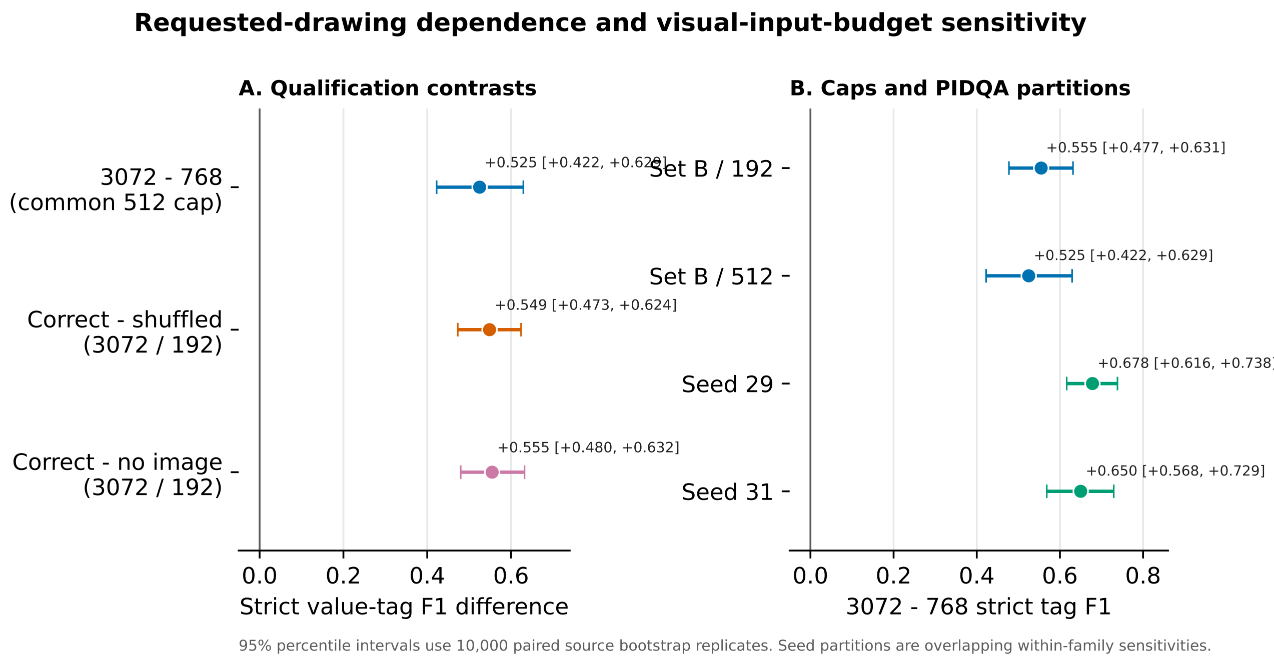

Figure 2: Paired task-specific qualification effects. Panel A contrasts correctly paired, source-shuffled, and absent image evidence. Panel B compares visual-input budgets under two output caps and two descriptive source partitions. Intervals use 10,000 paired source-cluster bootstrap replicates.

4.3 The requested-drawing effect replicates across scales, prompts, and subsets

The frozen extension contains 36 correct-minus-control comparisons. All 36 source-bootstrap intervals have positive lower bounds. Across the six subset–prompt cells, Qwen3-VL-8B correct-image value-tag F1 ranges from 0.5253 to 0.6587 (median 0.6229); correct-minus-shuffled effects range from +0.5191 to +0.6437, and correct-minus-no-image effects from +0.5253 to +0.6587. Qwen3-VL-32B correct-image F1 ranges from 0.5697 to 0.7361 (median 0.6512); its corresponding effect ranges are +0.5639–+0.7284 and +0.5697–+0.7361. Thus, the requested-drawing signal is not confined to one Qwen scale, subset, or wording.

InternVL3.5-8B also has positive correct-minus-control intervals in all 12 comparisons, but correct-image F1 remains 0.0090–0.0257 (median 0.0124). The direction therefore transfers more broadly than the useful magnitude. Eight of nine P1-minus-P0 intervals include zero; the exception is a small negative effect for Qwen3-VL-32B on seed 29. Figure 3 reports every pre-specified contrast rather than selecting a prompt or subset.

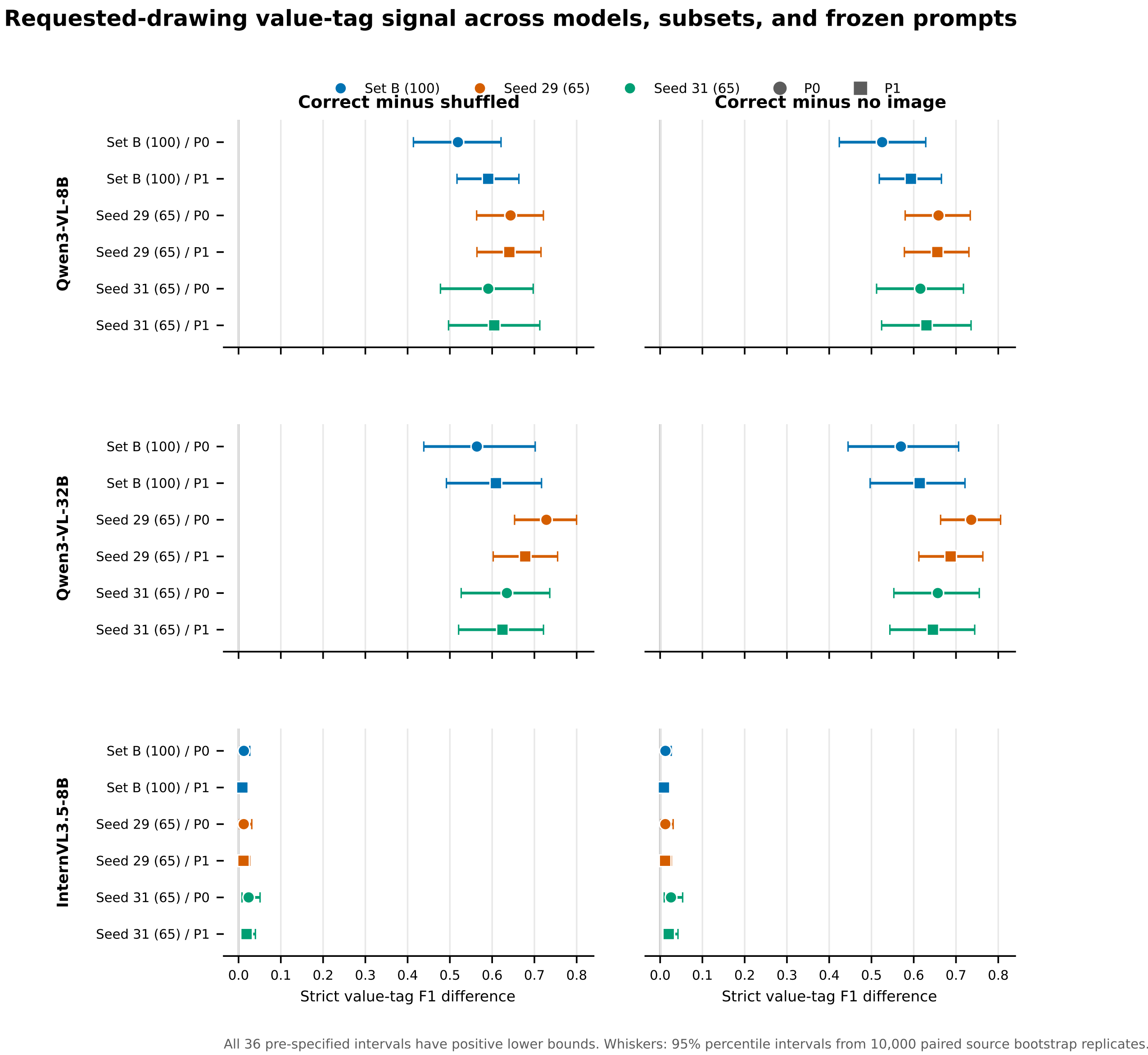

Figure 3: Frozen cross-model counterfactual replication. Rows cross three models with three pairwise source-disjoint PIDQA subsets and two pre-existing prompts. Columns show correct-image minus source-shuffled and correct-image minus no-image strict value-tag F1. Whiskers are 95% percentile intervals from 10,000 paired source bootstrap replicates.

4.4 OCR and VLM outputs support distinct operating modes

Table 2 reports counts and both F1 estimators. Geometry- joined OCR has fewer false positives than Qwen and pooled F1 0.5933. Union recovers 245 of 348 reference tags, increasing recall to 0.7040 and pooled F1 to 0.6339. Its pooled union-minus-Qwen difference is +0.0790 ([0.0372, 0.1261]); the source-macro difference is +0.0792 ([0.0373, 0.1250]). Non-empty-reference source-macro F1 is 0.6991 for union and 0.6139 for Qwen.

Intersection produces only one false candidate among 107 candidates: precision 0.9907 with source-bootstrap interval [0.9681, 1.0000], at recall 0.3046. OCR-first fallback reaches pooled F1 0.6228 and exact-set accuracy 0.2200. These are transparent retrieval modes, not a learned ensemble.

Table 2: Set B candidate-tag performance on 100 sources. Micro values pool TP/FP/FN; macro F1 weights each source equally.

Mode	TP	FP	FN	Precision	Recall	Micro F1	Macro F1	Exact
Qwen, correct image, 3072/192	192	152	156	0.5581	0.5517	0.5549	0.5810	0.1900
OCR, literal parse	116	44	232	0.7250	0.3333	0.4567	0.4227	0.1200
OCR, geometry join	159	29	189	0.8457	0.4569	0.5933	0.5618	0.2100
Set union	245	180	103	0.5765	0.7040	0.6339	0.6602	0.1900
Set intersection	106	1	242	0.9907	0.3046	0.4659	0.4312	0.1700
OCR if non-empty, else Qwen	175	39	173	0.8178	0.5029	0.6228	0.6073	0.2200

The median candidates per drawing [IQR] are 3 [1, 4] for Qwen, 2 [1, 3] for joined OCR, 4 [2, 5] for union, 1 [0, 2] for intersection, and 2 [1, 3] for OCR-first fallback. Median false candidates are respectively 0 [0, 2], 0 [0, 0], 1 [0, 2], 0 [0, 0], and 0 [0, 1]. Intersection is empty on 41 drawings, while union, Qwen, and OCR-first are empty on one each. This quantifies the coverage–review-burden trade-off behind Figure 4.

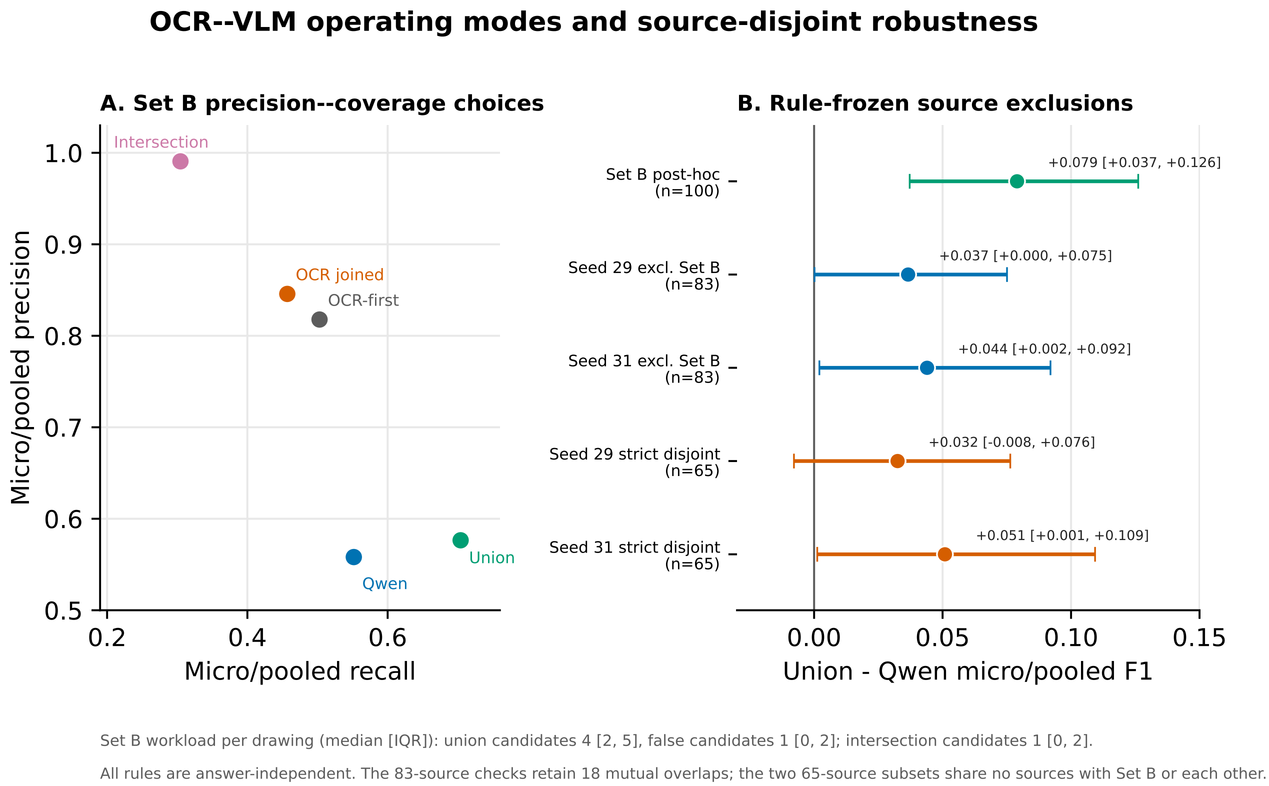

Figure 4: Reference-free operating modes and overlap-audited stability checks. Panel A gives Set B precision–recall locations. Panel B shows pooled union-minus-Qwen F1 for Set B, Set-B-excluded 83-source subsets, and mutually disjoint 65-source subsets. The footer reports per-drawing candidate workload.

4.5 Geometry joining and error complementarity explain the operating envelope

The reference-free OCR geometry rule performs 45 prefix–suffix joins. Relative to literal parsing, it adds 44 true tags and one false candidate while removing 16 false candidates and one true tag. Per-source F1 improves on 16 drawings, degrades on one, and is unchanged on 83. Pooled F1 increases from 0.4567 to 0.5933. This ablation supports the deterministic join as an engineering text normalization step rather than a reference-aware correction.

Of 348 true tags, 106 are recovered by both instruments, 86 only by Qwen, 53 only by OCR, and 103 by neither. Of 180 distinct false candidates appearing in their union, only one appears in both; 151 are Qwen-only and 28 OCR-only. At source level, Qwen contributes a unique true tag on 43 drawings, OCR on 17, and both contribute unique true tags on seven. The deterministic error taxonomy further shows that Qwen has 19 exact-set sources and 38 partial- recovery sources with false candidates, whereas joined OCR has 21 exact-set sources, 15 partial-recovery sources with false candidates, and 12 empty predictions against non-empty references. The full taxonomy and string-form diagnostics are reported in the supplement without causal interpretation.

4.6 Overlap-audited frozen-rule checks retain the direction with asymmetric strength

Set B overlaps seed 29 and seed 31 by 17 sources each; seed 29 and seed 31 overlap each other by 20 sources, including two shared by all three sets. Removing Set B leaves 83 sources per partition, with 18 still shared between the two partitions. On these Set-B-excluded subsets, pooled union-minus-Qwen F1 is +0.0366 ([0.0002, 0.0751]) and +0.0440 ([0.0021, 0.0920]). Source-macro effects are +0.0451 ([0.0043, 0.0892]) and +0.0373 ([-0.0082, 0.0854]).

Removing the residual cross-partition overlap yields 65 sources per strict subset. Pooled differences are +0.0325 ([-0.0078, 0.0764]) and +0.0509 ([0.0012, 0.1093]); source-macro differences are +0.0379 ([-0.0088, 0.0885]) and +0.0425 ([-0.0123, 0.0974]). The direction is positive in every point estimate, but interval exclusion of zero depends on subset and estimator. This supports a cautious within-family stability statement, not an independent-replication claim.

Table 3: Frozen union rule after source-overlap exclusions. Intervals are paired 95% source-bootstrap intervals.

Subset	n	Qwen micro F1	Union micro F1	Difference [95% interval]
Seed 29 excluding Set B	83	0.6717	0.7083	+0.0366 [0.0002, 0.0751]
Seed 31 excluding Set B	83	0.6417	0.6856	+0.0440 [0.0021, 0.0920]
Seed 29 strictly disjoint	65	0.6587	0.6911	+0.0325 [-0.0078, 0.0764]
Seed 31 strictly disjoint	65	0.6158	0.6667	+0.0509 [0.0012, 0.1093]

4.7 Boundary analyses delimit the supported claim

The qualification does not extend to topology or arbitrary model families. At the same 3072/192 Qwen operating point, correct-image aggregate strict accuracy is 0.2875 and no-image accuracy is 0.3025 because structural tasks retain strong prior pathways. A visible symbol legend changes spatial-count accuracy, but correct numeric labels show no detected advantage over a layout-matched label permutation (interval [-0.04, 0.04]). The initial corrected InternVL3.5-8B check produces correct-image tag F1 0.0126 and zero under source-shuffled and no-image conditions. The larger frozen matrix confirms a positive requested-drawing direction for InternVL but bounds its correct-image F1 at 0.0090–0.0257, far below the Qwen ranges. These results prevent the Qwen tag outcome from being generalized to a universal visual-input law, topology reasoning, or cross-model invariance.

5 Discussion

5.1 Engineering interpretation

The principal result is a bounded qualification: high-detail Qwen inference provides image-grounded candidate-tag retrieval on unseen PIDQA drawings. Correct/source-shuffled/no-image interventions localize the signal to the requested drawing, and the matched-output-cap comparison shows that the effect is not explained by a larger text-generation allowance. The aggregate score would have obscured this capability because nonvisual pathways dominate other task families. Reporting by task is therefore not merely presentational; it changes the engineering decision.

The OCR–VLM operating envelope turns the qualification into selectable workflow modes. Union emphasizes coverage (recall 0.7040) at a median of four candidates and one false candidate per drawing. Intersection yields an approximately one-candidate conservative shortlist with precision 0.9907 but is empty on 41% of drawings. OCR-first fallback provides a middle operating point with pooled F1 0.6228, a median of two candidates, and median zero false candidates. These counts allow a deployment owner to choose an operating mode according to missed-tag cost and downstream verification capacity.

5.2 Why the evaluation contract matters

SABER-PID joins four controls that are often considered separately. Drawing-level splitting removes an input-recoverable same-source route. Answer-hidden manifests separate inference from scoring. Counterfactual images test source identity and image presence directly. Recorded visual and output budgets make the high–low comparison interpretable. Finally, pooled and source-macro estimates show that conclusions are not an artifact of drawings with many tags.

The phase distinction is equally important. The Set B fusion result is a descriptive analysis formed after component inspection. Later rule checks freeze all operations, but their shared PIDQA family and source overlaps limit their evidential role. Explicit exclusions reveal the honest result: the positive direction persists, while uncertainty is sensitive to subset size and estimator. This is stronger and more useful than either suppressing the overlap or overstating the checks as independent evidence.

5.3 Implications for engineering document intelligence

The evaluated output is a candidate set for indexing or subsequent deterministic validation, not an autonomous engineering decision. This distinction aligns with practical P&ID digitization: text recognition is one layer of a larger pipeline that may incorporate format rules, master-data matching, and topology checks [3, 5]. The present results show when the image supplies usable tag evidence and how two frozen instruments trade coverage for precision. They do not remove the need for project-specific identifier rules or validation against controlled engineering data.

6 Limitations

PIDQA is one public synthetic 500-sheet source family. Source isolation protects unseen-sheet evaluation within that family but does not establish performance on scanned, revision-marked, proprietary, or project-specific plant drawings. The primary operating point concerns one frozen Qwen3-VL-8B prompt and decoding configuration. The extension covers two frozen prompts and a 32B Qwen scale, but the small InternVL3.5-8B effect shows that magnitude is not model-invariant. The OCR comparator is a frozen full-image pipeline rather than an optimized industrial recognizer.

Set B fusion is descriptive. Frozen-rule checks remain within PIDQA, and even after Set B exclusion the two 83-source subsets share sources; the stricter 65-source results have wider intervals. The 100-source controls estimate discrete operating points, not a continuous or universal visual-input law. Latency co-varies with generated output length and is not interpreted as pure image-encoding cost. Candidate workload is computed against benchmark references; field verification effort may differ.

An external PID2Graph/OPEN100 branch could not be evaluated because the obtained archive failed official size, MD5, and ZIP-central-directory checks. No member was extracted and no external score is reported. Cross-family validation therefore remains an external-validity need, not a condition silently treated as completed.

7 Conclusion

Candidate-tag retrieval from P&IDs can be qualified only when the evidence is reported at the task, source, model, and budget levels. On source-isolated PIDQA drawings, high-detail Qwen inference passes requested-drawing and matched-budget gates: pooled F1 is 0.5549 with the correct image, 0.0062 with a source-shuffled image, and zero without an image. A 54-cell frozen extension retains positive requested-drawing intervals in all 36 contrasts across two Qwen scales, InternVL, two prompts, and three source-disjoint subsets, while also exposing the much smaller InternVL magnitude. OCR and Qwen recover complementary tags. Reference-free union reaches recall 0.7040 and pooled F1 0.6339, while intersection reaches precision 0.9907. Overlap-audited frozen- rule checks preserve positive point estimates but also expose estimator- and subset-dependent uncertainty. The resulting conclusion is deliberately bounded: SABER-PID supports configurable candidate-tag retrieval within the evaluated PIDQA/Qwen setting; it does not qualify topology reasoning, general P&ID understanding, or field deployment.

Data and code availability

The submission package contains a locally validated reproducibility archive with answer-isolated public manifests, scorer-only references, immutable raw generation JSONL, deterministic analysis and figure scripts, tests, environment records, and SHA-256 manifests. PIDQA/Dataset-PID source data are covered by the vendored CC0 record; complete image collections are acquired by reference, and model weights are not redistributed. The submitter must upload this archive to a public repository and replace `[SUBMITTER: ARCHIVE DOI/URL]` before submission.

Declaration of competing interest

The submitting authors must insert and approve the applicable competing- interest declaration before upload; no declaration is inferred by the automated revision workflow.

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Use of generative AI and AI-assisted technologies

OpenAI Codex was used to assist analysis-code development, deterministic validation, figure generation, and manuscript editing. ChatGPT Pro was used for an author-side model-based editorial critique. All numerical statements in the manuscript were regenerated from machine-readable artifacts, and all figures were rendered by deterministic plotting code rather than a generative image model. The authors remain responsible for verification and the final submitted text.

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